(Please write your Enrollment No. immediately) Enrollment No. _____

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

6th Semester, March 2025

Paper Code: CIE 316 **Subject: Database Modelling and Design**

Time: 1½ Hrs. **Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	CO(s)
1(a)	What are different kinds of cardinalities ?	2	CO1
1(b)	Mention steps in database design life cycle.	2	CO1
1(c)	What is data independence?	2	CO1
1(d)	What is the role of foreign keys ?	2	CO2
1(e)	What is functional dependency?	2	CO2
2(a)	Explain the system structure of a database system with a neat block diagram.	5	Co1
2(b)	Compare and contrast between specialization and generalization process in Enhanced ER Modeling.	5	Co1
3(a)	Explain in detail about various key constraints used in the database system.	5	CO2
3(b)	Why Normalization is important.Given a relation R(P, Q, R, S, T, U, V, W, X, Y) and Functional Dependency set FD = { PQ → R, PS → VW, QS → TU, P → X, W → Y }, determine whether the given R is in 2NF? If not convert it into 2 NF.	5	CO2
4(a)	Draw an ER diagram for the Railway Reservation System.	5	CO1
4(b)	How ER model constructs can be mapped to the relational model?	5	CO2